Chapter II.2

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2.1. Let A be a ring, $X = \operatorname{Spec} A$ its spectrum. Fix an element $f \in A$, $D(f) = V(f)^c \subset X$ the associated distinguished open set. Show that $(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f$ as locally ringed spaces.

Proof. Let $Y = \operatorname{Spec} A_f$, $S = \{f^n\}_{n \geqslant 1}$, and $\phi : A \to S^{-1}A = A_f$ be the natural map $a \mapsto a/1$. Note that $f^n \in \mathfrak{p}$ prime for some $n \geqslant 1$ iff $f \in \mathfrak{p}$—that is, $S \cap \mathfrak{p} = \emptyset$ iff $\mathfrak{p} \notin D(f)$. Thus, AM 3.11.iv gives us a bijection

$$\begin{aligned} D(f) &\longleftrightarrow Y \\ \mathfrak{p} &\longrightarrow S^{-1}\mathfrak{p}, \\ \phi^{-1}(\mathfrak{q}) &\longleftarrow \mathfrak{q}. \end{aligned}$$

Clearly the map $\mathfrak{q} \mapsto \phi^{-1}(\mathfrak{q})$ is continuous, so it suffices to show that $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ is continuous.

Let $V(\mathfrak{b}) \subset A_f$ be closed, and denote $\psi : \mathfrak{p} \mapsto S^{-1}\mathfrak{p}$. AM 3.11.i gives us that $\mathfrak{b} = S^{-1}\mathfrak{a}$ for some ideal $\mathfrak{a} \trianglelefteqslant A$. Then $\mathfrak{p} \supset \mathfrak{a}$ implies $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a} = \mathfrak{b}$, so that $V(\mathfrak{a}) \cap D(f) \subset \psi^{-1}(V(\mathfrak{b}))$. Similarly, $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a}$ implies $\phi^{-1}(S^{-1}\mathfrak{p}) \supset \phi^{-1}(S^{-1}\mathfrak{a}) \supset \mathfrak{a}$. We conclude that $\psi^{-1}(V(\mathfrak{b})) = V(\mathfrak{a}) \cap D(f) \subset D(f)$ is closed, so that $\psi = (\psi^*)^{-1}$ is continuous.

By the above, $\phi : A \to A_f$ induces a map $(g, g^\sharp) : (\operatorname{Spec} A_f, \mathcal{O}_{\operatorname{Spec} A_f}) \to (D(f), \mathcal{O}_X|_{D(f)})$ which is a homeomorphism on underlying topological spaces. For each $\mathfrak{q} \not\ni f$,

$$g^\sharp_{\mathfrak{q}} : \mathcal{O}_{Y,\mathfrak{q}} \cong (A_f)_{\mathfrak{q}} \xrightarrow{\cong} (A_{g(\mathfrak{q})})_f \cong (\mathcal{O}_X|_{D(f)})_{g(\mathfrak{q})}$$

is an isomorphism, so $g^\sharp$ is an isomorphism of sheaves. Hence, $(g, g^\sharp)$ is an isomorphism of schemes. $\square$

2.2. Let $(X, \mathcal{O}_X)$ be a scheme, and let $U \subset X$ be any open subset. Show that $(U, \mathcal{O}_X|_U)$ is a scheme. We call this the *induced scheme structure* on the open set U, and we refer to $(U, \mathcal{O}_X|_U)$ as the open subscheme of U.

Proof. Choose a basic cover $\{D(f_i)\}$ of U. Then each

$$(D(f_i), (\mathcal{O}_X|_U)_{D(f_i)}) = (D(f_i), \mathcal{O}_X|_{D(f_i)}) \cong \operatorname{Spec} A_{f_i}$$

by 2.1. We conclude that $\{D(f_i)\}$ is an affine open cover of $(U, \mathcal{O}_X|_U)$, so that the latter locally ringed space is a sheaf. $\square$

2.3. $(*)$ A scheme $(X, \mathcal{O}_X)$ is *reduced* if for every open set $U \subset X$, the ring $\mathcal{O}_X(U)$ has no nilpotent elements.

(a) Show that $(X, \mathcal{O}_X)$ is reduced if and only if for every $x \in X$, the local ring $\mathcal{O}_{X,x}$ as no nilpotent elements.

(b) Let $(X, \mathcal{O}_X)$ be a scheme. Let $(\mathcal{O}_X)_{\mathrm{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\mathrm{red}}$, where for any ring A, we denote by A_{red} the quotient of A by its nilradical $\mathfrak{N}(A)$. Show that $(X, (\mathcal{O}_X)_{\mathrm{red}})$ is a scheme. We call it the *reduced scheme* associated to X, and denote it by X_{red}. Show that there is a morphism of schemes $X_{\mathrm{red}} \to X$, which is a homeomorphism on the underlying topological spaces.

(c) Let $f : X \to Y$ be a morphism of schemes, and assume that X is reduced. Show that there is a unique morphism $g : X \to Y_{\mathrm{red}}$ such that f is obtained by composing g with the natural map $Y_{\mathrm{red}} \to Y$.

Proof. (a) If $(X, \mathcal{O}_X)$ is not reduced, then we can take $U \neq \emptyset$ and $s \in \mathcal{O}_X(U)$ such that $s^n = 0$, whence $s_p^n = 0$ for all $x \in U$. Suppose conversely that there is $x \in X$ such that $\mathcal{O}_{X,x}$ has nilpotent elements. Take such a $(s, U) \in \mathcal{O}_{X,x}$ with $(s, U)^n = 0$. Then there is $V \supset U \supset p$ such that $(s|_V)^n = s^n|_V = 0$.

(b) Consider first the case of $X = \operatorname{Spec} A$ affine. Denote by $(\widetilde{\mathcal{O}_{\operatorname{Spec} A}})_{\mathrm{red}}$ the presheaf $U \mapsto \mathcal{O}_{\operatorname{Spec} A}(U)_{\mathrm{red}}$, so that $(\mathcal{O}_{\operatorname{Spec} A})_{\mathrm{red}} = (((\widetilde{\mathcal{O})_{\operatorname{Spec} A}})_{\mathrm{red}})^+$. By (AM, 3.11), there are isomorphisms $(A_{\mathrm{red}})_f \xrightarrow{\cong} (A_f)_{\mathrm{red}}$ whenever $f \notin \mathfrak{R}(A)$. The same proposition gives us $A_f = 0$ iff $f \in \mathfrak{p}$ for all primes $\mathfrak{p}$, i.e. $f \in \mathfrak{R}(A)$, in which case $(A_{\mathrm{red}})_f = 0$ as well. Then there is an isomorphism of presheaves $\phi : \mathcal{O}_{\operatorname{Spec}(A_{\mathrm{red}})} \to (\widetilde{\mathcal{O}_{\operatorname{Spec} A}})_{\mathrm{red}}$ given on a the distinguished affine opens by $(A_{\mathrm{red}})_f \xrightarrow{\cong} (A_f)_{\mathrm{red}}$. Hence, $(\widetilde{\mathcal{O}_{\operatorname{Spec} A}})_{\mathrm{red}}$ is itself a sheaf, and

$$(\operatorname{Spec} A)_{\mathrm{red}} \cong \operatorname{Spec} A_{\mathrm{red}}$$

is an affine scheme.

(2.3b) gives a natural map of affine schemes

$$(f, f^\sharp) : \operatorname{Spec} A_{\mathrm{red}} \to \operatorname{Spec} A$$

induced by the natural projection $\pi : A \twoheadrightarrow A_{\mathrm{red}}$. The map $f : \operatorname{Spec} A_{\mathrm{red}} \to \operatorname{Spec} A$ of topological spaces is a homeomorphism onto its image $V(\mathfrak{R}(A)) = \operatorname{Spec} A$, as desired.

Now, take X to be an arbitrary scheme, and $\operatorname{Spec} A \cong U \subset X$ an open affine neighborhood. We have

$$\begin{aligned}(U, \mathcal{O}_{X_{\mathrm{red}}}|_U) &\cong (\operatorname{Spec} A, \mathcal{O}_{(\operatorname{Spec} A)_{\mathrm{red}}}) \\ &\cong (\operatorname{Spec}(A_{\mathrm{red}}), \mathcal{O}_{\operatorname{Spec}(A_{\mathrm{red}})})\end{aligned}$$

is affine, so that X_{red} is a scheme. It suffices to supply a map $X_{\mathrm{red}} \to X$ which is a homeomorphism on topological spaces.

Take an open cover $\{\operatorname{Spec} A_i \cong U_i \subset X\}_{i \in I}$ of X. The affine case gives us maps $\varphi_i : \operatorname{Spec}(A_i)_{\mathrm{red}} \to \operatorname{Spec} A_i$. For $i, j \in I$, the affine communication lemma gives a cover of U_{ij} by $D(f_k)$ which are distinguished affine in both A_i, A_j. Since $\mathcal{O}_X(D(f_k)) \cong (A_i)_{f_k} \cong (A_j)_{f_k}$, the restrictions

$$\begin{aligned}\varphi_i|_{D(f_k)} &: D(f_k)_{\mathrm{red}} \to D(f_k), \\ \varphi_j|_{D(f_k)} &: D(f_k)_{\mathrm{red}} \to D(f_k)\end{aligned}$$

are induced by the same ring map $\mathcal{O}_X(D(f_k)) \twoheadrightarrow \mathcal{O}_X(D(f_k))_{\mathrm{red}}$, and are hence equal by (2.3). We proceed by making use of the following lemma.

Lemma. *For any schemes $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$, $U \mapsto \operatorname{Hom}_{\mathfrak{Sch}}((U, \mathcal{O}_X|_U), (Y, \mathcal{O}_Y))$ defines a sheaf of sets.*

Proof. Fix a cover U_i of X, and take maps $(f_i, f_i^\sharp) : (U_i, \mathcal{O}_X|_{U_i}) \to (Y, \mathcal{O}_Y)$ such that $(f_i, f_i^\sharp)|_{U_{ij}} = (f_j, f_j^\sharp)_{U_{ij}}$. That is, $f_i|_{U_{ij}} = f_j|_{U_{ij}}$, and the sheaf maps

$$f_i^\sharp|_{U_{ij}}, f_j^\sharp|_{U_{ij}} : U_{ij} \to (f_i|_{U_{ij}})_* \mathcal{O}_Y$$

coincide. It is elementary to show that $U \mapsto \mathrm{Hom}_{\mathsf{Top}}(U, Y)$ is a sheaf of sets, whence the $f_i : U_i \to Y$ glue to a unique continuous map $f : X \to Y$. By (Ex. 1.15), $\mathcal{H}om(\mathcal{O}_X, f_*\mathcal{O}_Y)$ is a sheaf, so the $f_i^\sharp : \mathcal{O}_X|_{U_i} \to (f_i)_*\mathcal{O}_Y$ glue to a unique sheaf morphism $f^\sharp : O_X \to f_*\mathcal{O}_Y$. The morphism $f_x^\sharp : \mathcal{O}_{X,x} \to (f_*\mathcal{O}_Y)_x$ is determined locally, and is thus a local morphism by locality of the f_i. Hence, our morphisms of locally ringed spaces $(f_i, f_i^\sharp)$ glue to a unique morphism of locally ringed spaces $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, as desired. $\square$

Composing of the inclusion $D(f_k) \hookrightarrow U_{ij}$, uniqueness of gluing implies that $\varphi_i|_{U_{ij}} : (U_{ij})_{\mathrm{red}} \hookrightarrow U_{ij}$ and $\varphi_j|_{U_{ij}} : (U_{ij})_{\mathrm{red}} \hookrightarrow U_{ij}$ coincide. Composing our $\varphi_i : \operatorname{Spec}(A_i)_{\mathrm{red}} \to \operatorname{Spec} A_i$ with the inclusion $\operatorname{Spec} A_i \hookrightarrow Y$, existence supplies us with our natural sheaf map $\varphi : X_{\mathrm{red}} \to X$, as desired.

Let $\psi_i := \varphi_i^{-1} : \operatorname{Spec} A_i \to \operatorname{Spec}(A_i)_{\mathrm{red}}$ be the inverse maps on the underlying topological spaces. Then

$$\psi_i|_{U_{ij}} = (\varphi_i|_{U_{ij}})^{-1} = (\varphi_j|_{U_{ij}})^{-1} = \psi_j|_{U_{ij}}$$

for all $i, j \in I$, so that the ψ_i glue to a continuous map $\psi : \operatorname{Spec} A \to (\operatorname{Spec} A)_{\mathrm{red}}$. Then

$$(\varphi \circ \psi)|_{U_i} = \psi_i \circ \phi_i = \mathrm{id}_{\operatorname{Spec} A_i}$$

for all $i \in I$ implies that $\varphi \circ \psi = \mathrm{id}_X$, and similarly we conclude $\psi \circ \varphi = \mathrm{id}_{X_{\mathrm{red}}}$. Hence, $\varphi : X_{\mathrm{red}} \to X$ is a homeomorphism on the underlying topological spaces.

(c) Suppose first that $Y \cong \operatorname{Spec} A$ is affine. Then (Ex. 2.4) gives a bijection between scheme maps $f : X \to \operatorname{Spec} A$ and ring maps $\varphi := \Gamma(X, f^\sharp) : A \to \Gamma(X, \mathcal{O}_X)$. Note that the natural map $\operatorname{Spec} A_{\mathrm{red}} \hookrightarrow \operatorname{Spec} A$ corresponds to the projection $\pi : A \twoheadrightarrow A_{\mathrm{red}}$. X reduced gives us $\varphi(\mathfrak{R}(A)) \subset \mathfrak{R}(\Gamma(X, \mathcal{O}_X)) = 0$, whence the universal property of quotients supplies us with a unique morphism $\psi : A_{\mathrm{red}} \to \Gamma(X, \mathcal{O}_X)$ such that $\psi \circ \pi = \varphi$. Under the correspondence, this ψ corresponds to a unique map $g : X \to \operatorname{Spec} A_{\mathrm{red}}$ such that the diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "g"'] & \operatorname{Spec} A \\ & \operatorname{Spec} A_{\mathrm{red}} \arrow[u, hook] \end{tikzcd}$$

commutes, as desired.

Let $f : X \to Y$, take a cover $\{U_i \cong \operatorname{Spec} A_i\}$ of Y, and let $V_i := f^{-1}(\operatorname{Spec} A_i)$. Then for each $f_i := f|_{V_i} : V_i \to \operatorname{Spec} A_i$, there is a unique scheme morphism $g_i : V_i \to \operatorname{Spec}(A_i)_{\mathrm{red}}$ making the following diagram commute.

$$\begin{tikzcd} V_i \arrow[r, "f_i"] \arrow[dr, "g_i"'] & \operatorname{Spec} A_i \\ & \operatorname{Spec}(A_i)_{\mathrm{red}} \arrow[u, hook] \end{tikzcd}$$

Cover U_{ij} by distinguished affines $D(a_k)$. The maps $g_i|_{D(f^{\sharp *}(a_k))}$ and $g_j|_{D(f^\sharp(a_k))}$ satisfy the given universal property for $f|_{D(f^\sharp(a_k))} : D(f^\sharp(a_k)) \to D(a_k)$, where $D(a_k) \cong \operatorname{Spec}(A_i)_{a_k}$ is affine. Hence, they are equal by the first case.

Composing with our inclusion, we have unqiue maps $g_i' : V_i \to \operatorname{Spec}(A_i)_{\mathrm{red}} \hookrightarrow Y_{\mathrm{red}}$ which restrict to the g_i. The proceeding lemma allows us to glue our g_i' to a unique map $g : X \to Y_{\mathrm{red}}$ such that $g|_{V_i} = g_i : V_i \to \operatorname{Spec}(A_i)_{\mathrm{red}}$. Let $j_i : \operatorname{Spec}(A_i)_{\mathrm{red}} \hookrightarrow \operatorname{Spec} A_i$ and $j : Y_{\mathrm{red}} \hookrightarrow Y$ be the natural maps. If $g' : X \to Y_{\mathrm{red}}$ has $f = j \circ g'$, then $f_i = j_i \circ g'|_{V_i}$ implies $g'|_{V_i} = g|_{V_i}$, whence $g' = g$.We conclude that $g : X \to Y_{\mathrm{red}}$ is in fact the unique map with this property, as desired. $\square$

2.4. Let A be a ring, and let $(X, \mathcal{O}_X)$ be a scheme. Given a morphism $f : X \to \operatorname{Spec} A$, we have an associated map $f^\sharp : \mathcal{O}_{\operatorname{Spec} A} \to f_* \mathcal{O}_X$ on sheaves. Taking global sections, we obtain a homomorphism $A \to \Gamma(X, \mathcal{O}_X)$. Thus, there is a natural map

$$\alpha : \operatorname{Hom}_{\mathfrak{Sch}}(X, \operatorname{Spec} A) \to \operatorname{Hom}_{\mathfrak{Ring}}(A, \Gamma(X, \mathcal{O}_X)).$$

Show that α is bijective.

Proof. We begin by showing the result for $X = \operatorname{Spec} B$ affine. In this case, α has an inverse map given by

$$\begin{aligned} \beta : \operatorname{Hom}_{\mathfrak{Ring}}(A, B) &\longrightarrow \operatorname{Hom}_{\mathfrak{Sch}}(\operatorname{Spec} B, \operatorname{Spec} A) \\ \varphi &\longmapsto (f, f^\sharp), \end{aligned}$$

where

$$\begin{aligned} f(\mathfrak{p}) &= \varphi^{-1}(\mathfrak{p}), \\ f^\sharp(s)(\mathfrak{p}) &= \varphi_{f(\mathfrak{p})}(s(f(\mathfrak{p}))). \end{aligned}$$

It suffices to show that β and α indeed compose to the identity.

Fix $(f, f^\sharp) \in \operatorname{Hom}_{\mathfrak{Sch}}(\operatorname{Spec} B, \operatorname{Spec} A)$, and denote

$$(\beta \circ \alpha)(f, f^\sharp) = (g, g^\sharp).$$

Let $\phi_{f(\mathfrak{p})} : A \to A_{f(\mathfrak{p})}$ and $\phi'_{\mathfrak{p}} : B \to B_{\mathfrak{p}}$ be the natural maps. The commutative square

$$\begin{tikzcd} A \arrow[r, "f^\sharp(\operatorname{Spec} A)"] \arrow[d, "\phi_{f(\mathfrak{p})}"'] & B \arrow[d, "\phi'_{\mathfrak{p}}"] \\ A_{f(\mathfrak{p})} \arrow[r, "f^\sharp_{f(\mathfrak{p})}"'] & B_{\mathfrak{p}} \end{tikzcd}$$

gives us that

$$\begin{aligned} g(\mathfrak{p}) &= (f^\sharp(\operatorname{Spec} A))^{-1}(\mathfrak{p}) \\ &= (\phi'_{\mathfrak{p}} \circ f^\sharp(X))^{-1}(m_{\mathfrak{p}}) \\ &= (f^\sharp_{f(\mathfrak{p})} \circ \phi_{f(\mathfrak{p})})^{-1}(m_{\mathfrak{p}}) \\ &= \phi^{-1}_{f(\mathfrak{p})}(m_{f(\mathfrak{p})}) \\ &= f(\mathfrak{p}), \end{aligned}$$

since $f^\sharp$ is a local homomorphism. Then

$$\begin{aligned} g^\sharp(s)(\mathfrak{p}) &= f^\sharp(X)_{g(\mathfrak{p})}(s(g(\mathfrak{p}))) \\ &= f^\sharp_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) \\ &= f^\sharp(s)(\mathfrak{p}). \end{aligned}$$

implies $(g, g^\sharp) = (f, f^\sharp)$.

Fix $\varphi : A \to B$, and denote $\beta(\varphi) = (f, f^\sharp)$, $(\alpha \circ \beta)(\varphi) = \psi$. Recall that $a \in A$ is identified with $\mathfrak{p} \mapsto a_\mathfrak{p} \in A_\mathfrak{p}$, and similarly for $b \in B$. Thus, $\psi(a) = \varphi(a)$ iff they're equal as maps $\operatorname{Spec} B \to \sqcup_{\mathfrak{p} \in \operatorname{Spec} B} B_\mathfrak{p}$. The commutative square

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d] & B \arrow[d] \\ A_{f(\mathfrak{p})} \arrow[r, "\varphi_{f(\mathfrak{p})}"'] & B_\mathfrak{p} \end{tikzcd}$$

gives us

$$\psi(a)(\mathfrak{p}) = \varphi_{f(\mathfrak{p})}(a(f(\mathfrak{p}))) = \varphi_{f(\mathfrak{p})}(a_{f(\mathfrak{p})}) = \varphi(a)(\mathfrak{p}).$$

Hence, $(\alpha \circ \beta)(\varphi) = \varphi$. We conclude that $\beta = \alpha^{-1}$, so α is a bijection.

Now, we consider the general case of X a scheme. Define a map

$$\begin{aligned} \beta : \operatorname{Hom}_{\mathfrak{Ring}}(A, \Gamma(X, \mathcal{O}_X)) &\longrightarrow \operatorname{Hom}_{\mathfrak{Sch}}(X, \operatorname{Spec} A) \\ \varphi &\longmapsto (f, f^\sharp) \end{aligned}$$

as follows. Fix an affine cover $\{U_i = \operatorname{Spec} B_i\}$ of X, and define $f : X \to \operatorname{Spec} A$ by $f(x) = \varphi^{-1}(\mathfrak{p}_x)$, where $\mathfrak{p}_x$ is the inverse image in $\Gamma(X, \mathcal{O}_X)$ of the unique maximal ideal $m_x \trianglelefteq (\mathcal{O}_X)_x$. Naturality of the restriction maps gives us that $f|_{U_i} = f_i$, where $(f_i, f_i^\sharp) = \beta(\rho_{U_i} \circ \varphi)$ is defined in the affine case.

Next, we define $f^\sharp$ as follows. Fix $V \subset X$ open and $s \in \mathcal{O}_{\operatorname{Spec} A}(V)$. Consider the sections $f_i^\sharp(V \cap U_i)(s) \in f_*\mathcal{O}_X(V \cap U_i)$, with $f_i^\sharp$ as above. We have that

$$f_i^\sharp(V \cap U_i \cap U_j)(s)(\mathfrak{p}) = (\rho_{V \cap U_i \cap U_j} \circ \varphi)_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) = f_j^\sharp(V \cap U_i \cap U_j)(s)(\mathfrak{p})$$

for all $\mathfrak{p} \in f^{-1}(V \cap U_i \cap U_j)$, so the sections agree on their overlap. Then they glue to a unique section $f^\sharp(V)(s) \in f_*\mathcal{O}_X(V)$.

It suffices to verify that α and β compose to the identity. Fix $(f, f^\sharp) \in \operatorname{Hom}_{\mathfrak{Sch}}(X, \operatorname{Spec} A)$, and denote $(\beta \circ \alpha)(f, f^\sharp) = (g, g^\sharp)$. Then $g|_{U_i} = g_i$, where

$$(g_i, g_i^\sharp) = \beta(\rho_{U_i} \circ f^\sharp(\operatorname{Spec} A)) = \beta(f^\sharp(U_i)).$$

This gives us $g_i = f|_{U_i}$ by the affine case, so that $g = f$.

Fix $V \subset \operatorname{Spec} A$ open and $s \in \mathcal{O}_{\operatorname{Spec} A}(V)$. Note that the $g^\sharp(V)(s)$ is the gluing of the sections $g^\sharp(V \cap U_i)(s)$ defined by

$$\begin{aligned} g^\sharp(V \cap U_i)(s)(\mathfrak{p}) &= (\rho_{V \cap U_i} \circ f^\sharp(V))_{g(\mathfrak{p})}(s(g(\mathfrak{p}))) \\ &= f^\sharp_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) \\ &= f^\sharp(V \cap U_i)(s)(\mathfrak{p}) \end{aligned}$$

for all $\mathfrak{p} \in f^{-1}(V \cap U_i)$. Thus, $(f, f^\sharp) = (g, g^\sharp)$.

Finally, fix $\varphi : A \to \Gamma(X, \mathcal{O}_X)$ and let $(f, f^\sharp) = \beta(\varphi)$, $\psi = (\alpha)(f, f^\sharp)$. Then

$$\rho_{U_i} \circ \psi = \rho_{U_i} \circ (\alpha(f, f^\sharp)) = f^\sharp(U_i) = \rho_{U_i} \circ \varphi$$

by the affine case, so that $\psi = \varphi$ by uniqueness of gluing. We conclude that α is a bijection. $\square$

2.5. Describe $\operatorname{Spec} \mathbb{Z}$, and show that it is a final object for the category of schemes, i.e., each scheme X admits a unique morphism to $\operatorname{Spec} \mathbb{Z}$.

Proof. The only primes of $\mathbb{Z}$ are (p) for p prime, and (0). The former are all maximal, while the latter is a generic point for $\operatorname{Spec}\mathbb{Z}$, since $V(\mathfrak{a}) \ni (0)$ implies $(0) \supset \mathfrak{a}$, i.e. $\mathfrak{a} = (0) \subset \mathfrak{p}$ for all $\mathfrak{p}$ prime.

We have a bijection

$$\operatorname{Hom}_{\mathfrak{Sch}}(X, \operatorname{Spec}\mathbb{Z}) \to \operatorname{Hom}_{\mathfrak{Ring}}(\mathbb{Z}, \Gamma(X, \mathcal{O}_X)),$$

so that the former is singleton for all X by the fact that $\mathbb{Z}$ is initial in $\mathfrak{Ring}$. $\square$

2.7. Let X be a scheme. For any point $x \in X$, let $\mathcal{O}_x$ be the local ring at x, and m_x its maximal ideal. We define the *residue field* of x on X to be the field $k(x) = \mathcal{O}_x/m_x$. Now, let K be any field. Show that to give a morphism of $\operatorname{Spec} K$ to X, it is equivalent to giving a point $x \in X$ and an inclusion $k(x) \rightarrowtail K$.

Proof. Note that $\operatorname{Spec} K = \{(0)\}$ is singleton, so a morphism $f : \operatorname{Spec} K \to X$ is a point $x = f((0)) \in X$, as well as the information $f^\sharp(U) : \mathcal{O}_X(U) \to K$ for every $x \in U \subset X$ open. There is a single non-empty open set $\{(0)\} = \operatorname{Spec} K\ \mathcal{O}_{\operatorname{Spec} K}$, so this gives us a map $f_x^\sharp : \mathcal{O}_{X,x} \to K$. Since $f_x^\sharp$ is a local morphism, we have ${f_x^\sharp}^{-1}(\mathfrak{m}_K) \subset \mathfrak{m}_{\mathcal{O}_{X,x}}$. Hence, $f_x^\sharp$ descends to a map $\overline{f_x^\sharp} : k(x) \to K$. It is a morphism of fields, and thus an injection.

Conversely, suppose we have an inclusion $k(x) \rightarrowtail K$. Then, for each $x \in U \subset X$, we have a map

$$f^\sharp(U) : \mathcal{O}(U) \to \mathcal{O}_{X,x} \twoheadrightarrow k(x) \rightarrowtail K.$$

These maps commute with restriction, thus giving us a morphism $f^\sharp : \mathcal{O}_X \to \mathcal{O}_{\operatorname{Spec} K}$ of sheaves. Defining $f : \operatorname{Spec} K \to X : (0) \mapsto x$ gives us a morphism $(f, f^\sharp) : \operatorname{Spec} K \to X$ of ringed spaces. It suffices to show that $f_x^\sharp : \mathcal{O}_{X,x} \to K$ is local. We have

$${f_x^\sharp}^{-1}(\mathfrak{m}_K) = \ker\left(\mathcal{O}_{X,x} \twoheadrightarrow k(x)\right) = \mathfrak{m}_{\mathcal{O}_{X,x}},$$

as desired. $\square$

2.9. If X is a topological space, and Z an irreducible closed subset of X, a generic point for Z is a point ζ such that $Z = \{\zeta\}^-$. If X is a scheme, show that every (nonempty) irreducible closed subset has a unique generic point.

Proof. We begin by proving the case of $X = \operatorname{Spec} A$ affine. Let $Z = V(\mathfrak{q}) \subset X$ be irreducible and closed. That is, $V(\mathfrak{q}) = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{ab})$ implies $V(\mathfrak{a}) = V(\mathfrak{q})$ or $V(\mathfrak{b}) = V(\mathfrak{q})$. Suppose now that $ab \in \mathfrak{q}$. Then $V((\mathfrak{q}, a)(\mathfrak{q}, b)) = V(\mathfrak{q})$ gives us $(\mathfrak{q}, a) \subset \sqrt{(\mathfrak{q}, a)} = \mathfrak{q}$ or $(\mathfrak{q}, b) \subset \sqrt{(\mathfrak{q}, b)} = \mathfrak{q}$. Hence, either $a \in \mathfrak{q}$ or $b \in \mathfrak{q}$, so that $\mathfrak{q}$ is prime.

Suppose that ζ is a generic point associated to $Z = V(\mathfrak{q})$. Then

$$\{\zeta\}^- = \bigcap_{V(\mathfrak{a}) \ni \zeta} V(\mathfrak{a}) = V(\cup_{\mathfrak{a} \subset \zeta} \mathfrak{a}) = V(\zeta)$$

implies that $\zeta = \mathfrak{q}$. This is clearly unique, and we showed that $\mathfrak{q}$ is prime above, so this is always possible.

Let Z be an irreducible closed subset of a scheme X, and let $\{U_i \cong \operatorname{Spec} A_i\}_{i \in I}$ be an affine open cover of X. Replace $\{U_i\}$ by a subcover such that each $U_i \cap Z \neq \emptyset$. We claim that each $U_i \cap Z$ is irreducible in U_i. Take $\emptyset \neq U_i \cap V \subset U_i \cap Z$ nonempty open in U_i. Then $\emptyset \neq U_i \cap V \subset Z$ nonempty open gives us that

$$\operatorname{Cl}_{U_i}(U_i \cap V) = U_i \cap \operatorname{Cl}_X(U_i \cap V) = U_i \cap Z,$$

whence $U_i \cap Z$ is irreducible in U_i.

Take $\zeta_i \in U_i$ to be the unique generic point of $U_i \cap Z$ given by the affine case, so $\{\zeta_i\}^- \supset \mathrm{Cl}_{U_i}(\zeta_i) = Z \cap U_i$. Note that $Z \cap U_i \cap U_j \neq 0$ for all i, j, as otherwise

$$Z = Z \cap (U_i^c \cup U_j^c) = (Z \cap U_i^c) \cup (Z \cap U_j^c)$$

gives us that Z is disjoint from U_i or U_j. Then

$$\{\zeta_i\}^- \cap U_j \supset (Z \cap U_i) \cap U_j \neq \emptyset$$

implies $\zeta_i \in U_j$, as otherwise $U_j^c \cap \{\zeta_i\}^- \supset \{\zeta_i\}^-$. Then $\mathrm{Cl}_{U_j}(\zeta_i) = \{\zeta_i\}^- \cap U_j$ implies

$$\mathrm{Cl}_{U_j}(\zeta_i) \supset (Z \cap U_i \cap U_j)^- \cap U_j = Z \cap U_j.$$

Hence, ζ_i is a generic point of $Z \cap U_j \subset U_j$ for each j. Uniqueness in the affine case gives us $\zeta_i = \zeta_j =: \zeta$ for all i, j, so that

$$\{\zeta\}^- \supset \bigcup_{i \in I} U_i \cap Z = Z.$$

Any generic point ζ' of Z must also be a generic point of each of the $U_i \cap Z$, so uniqueness follows from the affine case. □

2.13. A topological space is *quasi-compact* if every open cover has a finite sub-cover.

(a) Show that a topological space X is noetherian if and only if every open subset is quasi-compact.

(b) If A is a noetherian ring, show that $\mathrm{Sp}(\mathrm{Spec}\, A)$ is a noetherian topological space.

(c) Give an example to show that $\mathrm{Sp}(\mathrm{Spec}\, A)$ can be noetherian even when A is not.

Proof. □

2.14. (a) Let S be a graded ring. Show that $\mathrm{Proj}\, S = \emptyset$ if and only if every element of S_+ is nilpotent.

(b) Let $\varphi : S \to T$ be a graded homomorphism of graded rings (preserving degrees). Let $U = \{\mathfrak{p} \in \mathrm{Proj}\, T \ : \ \mathfrak{p} \not\supset \varphi(S_+)\}$. Show that U is an open subset of $\mathrm{Proj}\, T$, and show that φ determines a natural morphism $f : U \to \mathrm{Proj}\, S$.

(c) The morphism f can be an isomorphism even when φ is not. For example, suppose that $\varphi : S_d \to T_d$ is an isomorphism for all $d \geqslant d_0$, where d_0 is any integer. Then show that $U = \mathrm{Proj}\, T$ and the morphism $f : \mathrm{Proj}\, T \to \mathrm{Proj}\, S$ is an isomorphism.

Proof. (a) Denote $\mathrm{nil}'\, S := \cap_{\mathfrak{p}} \mathfrak{p}$, where $\mathfrak{p}$ ranges over all homogeneous prime ideals of S. It suffices to show that $\mathrm{nil}'\, S = \mathrm{nil}\, S$.

For any prime ideal $\mathfrak{p}$, the homogenization $h(\mathfrak{p}) \subset \mathfrak{p}$ generated by its homogeneous elements is clearly a homogeneous prime ideal. Then we have

$$\mathrm{nil}'\, S \supset \mathrm{nil}\, S = \bigcap_{\mathfrak{p}} \mathfrak{p} \supset \bigcap_{\mathfrak{p}} h(\mathfrak{p}) = \mathrm{nil}'\, S,$$

which gives us $\mathrm{nil}'\, S \subset \mathrm{nil}\, S$. We conclude that $\mathrm{nil}\, S = \mathrm{nil}'\, S$. Now, $\mathrm{Proj}\, S = \emptyset$ iff $S_+ \subset \mathrm{nil}'\, S = \mathrm{nil}\, S$, which is exactly our result.

(b) Let $\mathfrak{p} \in \mathrm{Proj}\, T$ be a homogeneous prime ideal. Then $\varphi^{-1}(\mathfrak{p})$ is a homogeneous prime ideal, and $\varphi^{-1}(\mathfrak{p}) \supset S_+$ implies

$$\mathfrak{p} \supset \varphi(\varphi^{-1}(\mathfrak{p})) \supset \varphi(S_+),$$

i.e. $\mathfrak{p} \notin U$. Conversely, $\mathfrak{p} \supset \varphi(S_+)$ implies

$$\varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\varphi(S_+)) \supset S_+,$$

so that $\varphi^{-1}(\mathfrak{p}) \notin \operatorname{Proj} S$. We conclude that U is the inverse image of $\operatorname{Proj} S$ under the mapping of homogeneous prime ideals $\mathfrak{p} \mapsto \varphi^{-1}$. Hence, we have a natural map $f : U \to \operatorname{Proj} S : \mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$.

Fix a homogeneous $g \in T_+$. Then

$$\varphi^{-1}(\mathfrak{p}) \ni g \implies \mathfrak{p} \supset \varphi(\varphi^{-1}(\mathfrak{p})) \ni \varphi(g) \implies \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\varphi(g)) \ni g$$

gives us that

$$f^{-1}(D_+(g)) = D_+(\varphi(g)) \subset U.$$

Hence, $f : U \to \operatorname{Proj} S$ is continuous. We must now define the map $f^\sharp : \mathcal{O}_S \to f_*(\mathcal{O}_T|_U)$ of sheaves.

It suffices to define $f^\sharp$ on basic open sets. Fix a homogeneous $g \in S_+$, and note that

$$f_*(\mathcal{O}_T|_U)(D_+(g)) = \mathcal{O}_T(f^{-1}(D_+(g)) \cap U) = \mathcal{O}_T(D_+(\varphi(g))) \cong T_{(\varphi(g))}.$$

Recall as that φ is degree preserving, so each of its localizations are as well. In particular, $\varphi_g : S_g \to T_g$ restricts to a map $\varphi_g : S_{(g)} \to T_{(\varphi(g))}$. Then the following commutative diagram gives us our sheaf map on $D_+(g)$.

$$\begin{tikzcd}
\mathcal{O}_S(D_+(g)) \arrow[r, "f^\sharp(D_+(g))"] \arrow[d, "\wr"] & f_*(\mathcal{O}_T|_U)(D_+(g)) \arrow[d, "\wr"] \\
S_{(g)} \arrow[r, "\varphi_g"'] & T_{(\varphi(g))}
\end{tikzcd}$$

To show that these maps glue together, we note that $D(g) \cap D(h) = D(gh)$, and consider the following commutative diagram.

$$\begin{tikzcd}
S_{(g)} \arrow[r, "\varphi_g"] \arrow[dr] & S_{(\varphi(g))} \arrow[dr] & \\
& S_{(gh)} \arrow[r, "\varphi_{gh}"] & S_{(\varphi(gh))} \\
S_{(h)} \arrow[r, "\varphi_h"'] \arrow[ur] & S_{(\varphi(h))} \arrow[ur] &
\end{tikzcd}$$

This defines our sheaf homomorphism on all of $\operatorname{Proj} S$, so that we're done.

(c) Fix d_0 such that $\varphi : S_d \to T_d$ is an isomorphism for all $d \geqslant d_0$. We wish to show that $U = \operatorname{Proj} T$, so it suffices to show that $\sqrt{\varphi(S_+)} = T_+$. One inclusion is clear since φ is degree preserving. Surjectivity of $\varphi : S_d \to T_d$ gives us $\varphi(S_+) \supset \oplus_{d \geqslant d_0} T_d$. Then, for any element $a \in T_+$, a^{d_0} is of degree at least d_0, so $a^{d_0} \in \varphi(S_+)$ implies $a \in \sqrt{\varphi(S_+)}$. We conclude that $\sqrt{\varphi(S_+)} = T_+$, so $U = \operatorname{Proj} T$.

It suffices to show that $f^\sharp$ is an isomorphism on open sets $D(g)$. Fix homogeneous $g \in S_+$. Since $D(g^n) = D(g)$ for all $n > 0$, we may assume g is of degree $\geqslant d_0$. We show the map $\varphi_g : S_{(g)} \to T_{(\varphi(g))}$ is surjective. For any $b/\varphi(g)^n \in (B_A)_{\mathfrak{p}}$, $b\varphi(g) = \varphi(a)$ with $a \in A$ gives us

$$\varphi_g(a/g^{n+1}) = \varphi(ag)/\varphi(g^{n+1}) = b/\varphi(g)^n \in \operatorname{im} \varphi_g.$$

It suffices to check injectivity. If $\varphi(a/g) = 0$, then we can find $n \geqslant 1$ such that $\varphi(g)^n\varphi(a) = 0$. Then $\varphi(g^n a) = 0$ implies $g^n a = 0$ by injectivity of φ, whence $a/g = 0$ in $S_{(g)}$. The result follows. $\square$

2.16. Let X be a scheme, let $f \in \Gamma(X, \mathcal{O}_X)$, and define

$$X_f := \{x \in X : f_x \notin \mathfrak{m}_x \subset \mathcal{O}_{X,x}\}.$$

(a) If $U = \operatorname{Spec} B$ is an open affine subscheme of X, and if $\overline{f} \in B = \Gamma(U, \mathcal{O}_X|_U)$ is the restriction of f, show that $U \cap X_f = D(\overline{f})$. Conclude that X_f is an open subset of X.
(b) Assume that X is quasi-compact. Let $A = \Gamma(X, \mathcal{O}_X)$, and let $a \in A$ be an element whose restriction to X_f is 0. Show that, for some $n > 0$, $f^n a = 0$.
(c) Now assume that X has a finite cover by open affines U_i such that each intersection $U_i \cap U_j$ is quasi-compact. (This hypothesis is satisfied, for example, if $\operatorname{sp}(X)$ is noetherian.) Let $b \in \Gamma(X_f, \mathcal{O}_{X_f})$. Show that, for some $n > 0$, $f^n b$ is the restriction of an element of A.
(d) With the hypothesis of (c), conclude that $\Gamma(X_f, \mathcal{O}_{X_f}) \cong A_f$.

Proof. (a) $\mathfrak{p} \in U \cap X_f$ iff $\overline{f}_{\mathfrak{p}} \notin \mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}\mathcal{O}_{U,\mathfrak{p}}$, i.e. $\overline{f} \notin \mathfrak{p}$. It follows that $U \cap X_f = D(\overline{f})$ is open, so that X_f is a union of open sets, hence open itself.

(b) Let $a \in A$ with $a|_{X_f} = 0$. Then, for any open affine U intersection X_f, we define denote the restriction $\overline{a} := a|_U$. Then $\overline{a}|_{D(\overline{f})} = a|_{U\cap X_f} = 0$, whence there exists $n > 0$ such that $\overline{f^n a} = \overline{f}^n\overline{a} = 0$. Quasi-compactness of X gives us a finite affine open cover $\{U_i\}$, so that we can take some fixed $n > 0$ with $(f^n a)|_{U_i} = 0$ for all U_i. The sheaf axiom gives us that in fact $f^n a = 0$, so we are done.

(c) Consider $f_i := f|_{U_i}$ and $b_i := b|_{D(f_i)}$. Then, for each i, (a) gives us some $m > 0$ such that

$$(f_i^m b_i)|_{D(f_i)} = c_i|_{D(f_i)},$$

with $c_i \in \Gamma(U_i, \mathcal{O}_{U_i})$. Denote $U_{ij} = U_i \cap U_j$. $(c_i - c_j)|_{U_{ij}\cap X_f} = 0$, so (c) gives us $l > 0$ such that $f^l(c_i - c_j)|_{U_ij} = 0$, i.e. $(f^l c_i)|_{U_{ij}} = (f^l c_j)|_{U_{ij}}$. There are only finitely many U_{ij}, so we can take l such that the above holds for each i, j. Then we have $n = m + l > 0$ such that

$$(f_i^n b_i)|_{D(f_i)} = c_i|_{D(f_i)},$$
$$c_i|_{U_ij} = c_j|_{U_{ij}}$$

for all i, j. Then the c_i glue to an element $c \in A$ such that $c|_{X_f}$ is the gluing of the $f_i^n b_i|_{D(f_i)}$; namely, $f^n b$.

(d) By (c), we have a map $\phi : \Gamma(X_f, \mathcal{O}_{X_f}) \to A_f : b \mapsto f^n b/f^n$, which is well defined regardless of our choice of $n > 0$.[1] Suppose that $\phi(b) = 0$, so that there exists $m > 0$ with $f^{n+m}b = f^m(f^n b \cdot 1 - f^n \cdot 0) = 0$. Then, with notation as in (c), $f_i^{n+m}b_i = 0$ means $b_i = 0$ for each i, whence $b = 0$, showing injectivity.

Finally, we show surjectivity. Fix $a/f^m \in A_f$. It suffices to find $b \in \Gamma(X_f, \mathcal{O}_{X_f})$ such that $f^{n+m}b - f^n a = 0$ for $f^n b \in A$.[2] Fortunately, we need only glue together the maps

[1] We are often somewhat sloppy in writing $f^n b$ for the element of A which restricts to it, but we attempt to distinguish when necessary.

[2] It is in fact equivalent, as if $f^{n+m+l}b - f^{n+l}a = 0$ for $l > 0$, then we can replace $n > 0$ by $n + l > 0$.

$b_i = a|_{D(f_i)}/f_i^m$ to a map $b \in \Gamma(X_f, \mathcal{O}_{X_f})$, where compatibility is simple. Then $f^m b = a$, so that $\phi(b) = f^m b/f^m = a/f^n$, as desired. We conclude that ϕ is an isomorphism of rings. $\square$

2.17. *A Criterion for Affineness*

(a) Let $f : X \to Y$ be a morphism of schemes, and suppose that Y can be covered by open subsets U_i such that for each i, the induced map $f^{-1}(U_i) \to U_i$ is an isomorphism. Then f is an isomorphism.

(b) A scheme X is affine if and only if there is a finite set of elements $f_1, \ldots, f_r \in A = \Gamma(X, \mathcal{O}_X)$ such that the open subsets X_{f_i} are affine, and $f_1, \ldots, f_r$ generate the unit ideal in A.

Proof. (a) Let $f_i := f|_{U_i} : f^{-1}(U_i) \to U_i$, and take inverses $g_i = f_i^{-1} : U_i \to f^{-1}(U_i)$. We have

$$g_i|_{U_{ij}} = f_i^{-1}|_{U_{ij}} = f_j^{-1}|_{U_{ij}} = g_j|_{U_{ij}},$$

so the maps agree on overlap. Hence, they glue to a unique map $g : Y \to X$ such that

$$(g \circ f)|_{f^{-1}(U_i)} = g_i \circ f_i = \mathrm{id}_{f^{-1}(U_i)},$$
$$(f \circ g)|_{U_i} = f_i \circ g_i = \mathrm{id}_{U_i}.$$

Then $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$ by uniqueness of gluing, so that $g = f^{-1}$. We conclude that f is indeed an isomorphism.

(b) If X is affine, then we can take $(f_1, \ldots, f_r) = (1)$ to be a finite set of generators of the unit ideal, so that $X_{f_i} = D(f_i)$ are distinguished affines covering X.

Conversely, take a scheme X with such elements $f_1, \ldots, f_r \in A$. Each X_{f_i} is affine and thus quasi-compact, so a finite cover of X by quasi-compact open sets implies X is quasi-compact. Each intersection $X_{f_i} \cap X_{f_j} = D_{X_{f_i}}(f_j) \cong \mathrm{Spec}(A_{f_i})_{f_j} \cong \mathrm{Spec}\, A_{f_i f_j}$ is affine with and thus quasi-compact, so that 2.16(d) gives us $\Gamma(X_{f_i}, \mathcal{O}_{X_{f_i}}) \cong A_{f_i}$.

Ex. 2.4 supplies us with an isomorphism $\phi_i : X_{f_i} \xrightarrow{\cong} \mathrm{Spec}\, A_{f_i}$, so that we have a cover of $\mathrm{Spec}\, A$ by $X_{f_i} \cong \mathrm{Spec}\, A_{f_i}$ where $\phi_i|_{D_{X_{f_i}}(f_j)} = \phi_j|_{D_{X_{f_j}}(f_i)}$. Then we can glue together the maps ϕ_i to a map $\phi : X \to \mathrm{Spec}\, A$, which is an isomorphism by (a). Hence, $X \cong \mathrm{Spec}\, A$ is affine. $\square$

2.18. In this exercise, we compare some properties of a ring homomorphism to the induced morphism of the spectra of the rings.

(a) Let A be a ring, $X = \mathrm{Spec}\, A$, and $f \in A$. Show that f is nilpotent if and only if $D(f)$ is empty.

(b) Let $\varphi : A \to B$ be a homomorphism of rings, and let $f : Y = \mathrm{Spec}\, B \to X = \mathrm{Spec}\, A$ be the induced morphism of affine schemes. Show that φ is injective if and only if the map of sheaves $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is injective. Show furthermore in that case f is *dominant*, i.e., $f(Y)$ is dense in X.

(c) With the same notation, show that if φ is surjective, then f is a homeomorphism of Y onto a closed subset of X, and $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective.

(d) Prove the converse to (c). Namely, if $f : Y \to X$ is a homeomorphism onto a closed subset, and $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective, then φ is surjective.

Proof. (a) The result is trivial, as $D(f) = \emptyset$ iff $f \in \cap_{\mathfrak{p}} \mathfrak{p} = \mathrm{nil}\, f$.

(b) In the language of coherent sheaves, we have $\mathcal{O}_X = \widetilde{A}$ and $f_*\mathcal{O}_Y = f_*\widetilde{B} = \widetilde{B_A}$. Then $f^\sharp_{\mathfrak{p}} : \mathcal{O}_{X,\mathfrak{p}} \to (f_*\mathcal{O}_Y)_{\mathfrak{p}}$ corresponds to the map $\varphi_{\mathfrak{p}} : A_{\mathfrak{p}} \to (B_A)_{\mathfrak{p}}$. By CA, we have that φ is injective iff each $\varphi_{\mathfrak{p}}$ is injective, which by the above occurs iff $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is injective.

It suffices to show that $f(Y)$ is dense in X. Fix $g \in A$. Then

$$\begin{aligned} f(Y) \cap D(g) = \emptyset &\iff f^{-1}(D(g)) = \emptyset \\ &\iff D(\varphi(g)) = \emptyset \\ &\iff \varphi(g) \in \operatorname{nil} B \\ &\iff g \in \operatorname{nil} A \\ &\iff D(g) = \emptyset \end{aligned}$$

by (a) and injectivity of φ respectively. Any nonempty open $U \subset X$ contains a nonempty distinguished open of X, so $f(Y)$ is dense in X.

(c) First, suppose that $\varphi : A \to B$ is surjective. Localization is exact, so each of the stalks $\varphi_{\mathfrak{p}} : A_{\mathfrak{p}} \to (B_A)_{\mathfrak{p}}$ is surjective, whence $f^\sharp : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective. Surjectivity of φ gives us

$$\mathfrak{p} \supset \mathfrak{a} \implies \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\mathfrak{a}) \implies \mathfrak{p} = \varphi(\varphi^{-1}(\mathfrak{p})) \supset \varphi(\varphi^{-1}(\mathfrak{a})) = \mathfrak{a},$$

so that

$$\begin{aligned} f(V(\mathfrak{a})) &= \{\varphi^{-1}(\mathfrak{p}) \in X \ : \ \mathfrak{p} \supset \mathfrak{a}\} \\ &= f(Y) \cap V(\varphi^{-1}(\mathfrak{a})). \end{aligned}$$

In particular,

$$f(Y) = f(Y) \cap V(\ker\varphi) \subset V(\ker\varphi).$$

Take $\mathfrak{q} \in V(\ker\varphi)$, and suppose there exists $a \in \varphi^{-1}(\varphi(\mathfrak{q})) \setminus \mathfrak{q}$. Then there is some $a' \in \mathfrak{q}$ with $\varphi(a) = \varphi(a')$, so that $a - a' \in \ker\varphi \setminus \mathfrak{q} = \emptyset$, a contradiction. Thus, $\varphi^{-1}(\varphi(\mathfrak{q})) = \mathfrak{q}$.

We now show that $\varphi(\mathfrak{q})$ is prime. Suppose $bb' \in \varphi(\mathfrak{q})$, and take $a, a' \in A$ such that $\varphi(a) = b$ and $\varphi(a') = b'$. Then $aa' \in \varphi^{-1}(\varphi(\mathfrak{q})) = \mathfrak{q}$ implies $a \in \mathfrak{q}$ or $a' \in \mathfrak{q}$. Hence, $b \in \varphi(\mathfrak{q})$ or $b' \in \varphi(\mathfrak{q})$ gives us that $\varphi(\mathfrak{q})$ is prime, so that $g : V(\ker\varphi) \to Y : \mathfrak{q} \to \varphi(\mathfrak{q})$ is a set-theoretic inverse to $f : Y \to V(\ker\varphi) \subset X$. In particular, $f(Y) = V(\ker\varphi)$.

It suffices to show that $g : V(\ker\varphi) \to Y$ is continuous. We write

$$f(V(\mathfrak{a})) = f(V) \cap V(\varphi^{-1}(\mathfrak{a})) = V(\ker\varphi + \varphi^{-1}(\mathfrak{a})),$$

so that $g^{-1}(V(\mathfrak{a})) = V(\ker\varphi + \varphi^{-1}(\mathfrak{a}))$ is closed in X. We conclude that f is a homeomorphism onto its closed image $f(Y) = V(\ker\varphi)$.

(d) Suppose now that $f : Y \to X$ is a homeomorphism onto a closed subset, and $f^\sharp : \mathcal{O}_X \to \mathcal{O}_Y$ is surjective. Let $Z = \operatorname{Spec}(A/\ker\varphi)$, so that the universal property of quotients supplies $\overline{\varphi} : A/\ker\varphi \to B$ such that $\varphi = \overline{\varphi} \circ \pi$. We denote the associated morphism of sheaves $\overline{f} : Y \to Z$. Note that $\pi : A \to A/\ker\varphi$ is surjective, so (c) gives us that the associated map $g : Z \to V(\ker\varphi) \subset X$ is a homeomorphism.

$\overline{\varphi}$ is injective, so (b) implies $\overline{f}^\sharp : \mathcal{O}_Z \to \mathcal{O}_Y$ is injective, and $\overline{f}(Y) = f(Y)$ is dense. $f(Y) = f(Y)^-$ is closed, so $\overline{f} : Y \to Z$ is surjective. $f = g \circ \overline{f}$ with f, g homeomorphisms onto their image implies $\overline{f}$ is a homeomorphism onto its image $\overline{f}(Y) = Z$. Additionally, $f^\sharp = \overline{f}^\sharp g^\sharp$ surjective implies $\overline{f}^\sharp$ is surjective, so that $\overline{f}^\sharp : \mathcal{O}_Z \to \mathcal{O}_Y$ is an isomorphism of sheaves. Then

$$\varphi = \overline{\varphi} \circ \pi = \overline{f}^\sharp(Z) \circ \pi$$

is the composition of a surjective map with an isomorphism, and is therefore surjective itself. $\square$